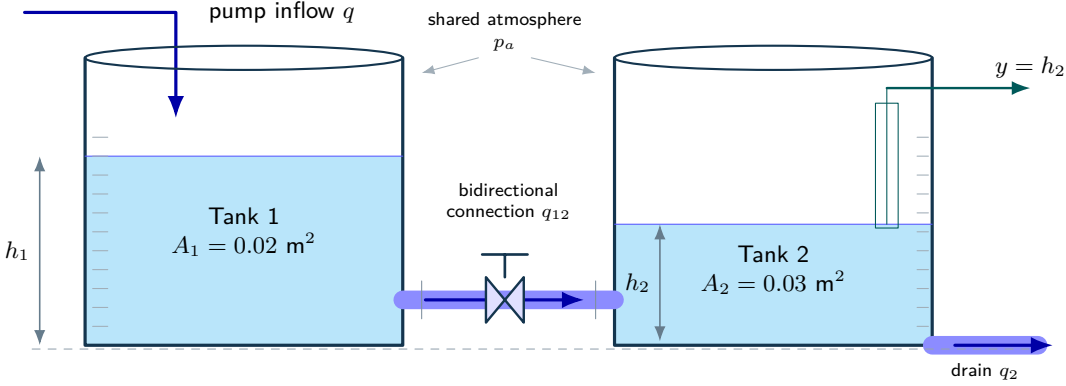

Common height datum; open tanks, constant areas, ideal gravity-flow laws

Two volume inventories

$A_1 \dot{h}_1 = q - q_{12}$.
$A_2 \dot{h}_2 = q_{12} - q_2$.
Only tank 2 drains to the exterior.

Gravity-driven coupling

$q_{12} = k_{12}\,\mathrm{sgn}(h_1 - h_2)$
$\cdot\sqrt{|h_1 - h_2|}$.
$q_2 = k_2\sqrt{h_2}$.

Interpret the arrows

Displayed levels give forward connection flow; reversal is allowed. The open rims share atmospheric pressure and one elevation datum.